Divide and Conquer – Searching & Basic Problems

Binary Search, Finding Maximum and Minimum using Divide and Conquer

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Question 1: Search an Element in a Sorted Array

Given a sorted array of integers and a key value, use the Binary Search (Divide and Conquer) method to determine whether the key is present in the array. If found, print its position; otherwise, print "Element not found".

Test Case 1

Input

5
10 20 30 40 50
30

Expected Output

Element found at position 3

Test Case 2

Input

5
10 20 30 40 50
25

Expected Output

Element not found

```
n=int(input())
a=list(map(int,input().split()))
k=int(input())
l,h=0,n-1
while l<=h:
    m=(l+h)//2
    if a[m]==k:
        print("Element found at position",m+1)
        break
    elif a[m]<k:
        l=m+1
    else:
        h=m-1
else:
    print("Element not found")
```

```
5
10 20 30 40 50
30
Element found at position 3
```

Question 2: Find the Index of a Given Number

Given a sorted array and a target number, find the index of the target using Binary Search. Assume array indexing starts from 0.

Test Case 1

Input

6
2 4 6 8 10 12
8

Expected Output

Index = 3

Test Case 2

Input

6
 2 4 6 8 10 12
 5

Expected Output

Index = -1

```python
n=int(input())
a=list(map(int,input().split()))
k=int(input())
l,h=0,n-1
while l<=h:
    m=(l+h)//2
    if a[m]==k:
        print("Index =",m)
        break
    elif a[m]<k:
        l=m+1
    else:
        h=m-1
else:
    print("Index = -1")
```

```
6
2 4 6 8 10 12
8
Index = 3
```

Question 3: Count the Number of Iterations in Binary Search

Given a sorted array and a key, perform Binary Search and display the number of iterations required to find the element.

Test Case 1

Input

7
 5 10 15 20 25 30 35
 25

Expected Output

Element found
 Iterations = 2

Test Case 2

Input

7
 5 10 15 20 25 30 35
 40

Expected Output

Element not found
 Iterations = 3

```
n=int(input())
a=list(map(int,input().split()))
k=int(input())
l,h=0,n-1
c=0
while l<=h:
    c+=1
    m=(l+h)//2
    if a[m]==k:
        print("Element found")
        print("Iterations =",c)
        break
    elif a[m]<k:
        l=m+1
    else:
        h=m-1
else:
    print("Element not found")
    print("Iterations =",c)
```

```
7
5 10 15 20 25 30 35
25
Element found
Iterations = 3
```

Question 4: Find the First Occurrence of an Element

Given a sorted array containing duplicate elements, find the first occurrence of a given key using Binary Search.

Test Case 1

Input

8
1 2 2 2 3 4 5 6
2

Expected Output

First occurrence at index 1

Test Case 2

Input

8
1 2 2 2 3 4 5 6
7

Expected Output

Element not found

```
n=int(input())
a=list(map(int,input().split()))
k=int(input())
l,h=0,n-1
ans=-1
while l<=h:
    m=(l+h)//2
    if a[m]==k:
        ans=m
        h=m-1
    elif a[m]<k:
        l=m+1
    else:
        h=m-1

if ans!=-1:
    print("First occurrence at index",ans)
else:
    print("Element not found")
```

```
8
1 2 2 2 3 4 5 6
2
First occurrence at index 1
```

Question 5: Find the Last Occurrence of an Element

Given a sorted array containing duplicate elements, find the last occurrence of a given key using Binary Search.

Test Case 1

Input

8
1 2 2 2 3 4 5 6
2

Expected Output

Last occurrence at index 3

Test Case 2

Input

8
1 2 2 2 3 4 5 6
9

Expected Output

Element not found

```
n=int(input())
a=list(map(int,input().split()))
k=int(input())
l,h=0,n-1
ans=-1
while l<=h:
    m=(l+h)//2
    if a[m]==k:
        ans=m
        l=m+1
    elif a[m]<k:
        l=m+1
    else:
        h=m-1
if ans!=-1:
    print("Last occurrence at index",ans)
else:
    print("Element not found")
```

```
8
1 2 2 2 3 4 5 6
2
Last occurrence at index 3
```

Question 6: Highest and Lowest Temperature Recorded

A weather station records temperatures over a week. Find the **minimum** and **maximum** temperature using the Divide and Conquer approach.

Input

7
32 28 35 25 31 29 37

Expected Output

Minimum Temperature = 25
Maximum Temperature = 37

```
def mm(a):
    if len(a)==1:
        return a[0],a[0]
    m=len(a)//2
    x1,y1=mm(a[:m])
    x2,y2=mm(a[m:])
    return min(x1,x2),max(y1,y2)
n=int(input())
a=list(map(int,input().split()))
mn,mx=mm(a)
print("Minimum Temperature =",mn)
print("Maximum Temperature =",mx)
```

```
7
32 28 35 25 31 29 37
Minimum Temperature = 25
Maximum Temperature = 37
```

Question 7: Best and Worst Student Marks

A teacher wants to identify the highest and lowest marks scored in a class using Divide and Conquer.

Input

8
78 92 65 88 95 72 81 69

Expected Output

Minimum Mark = 65
Maximum Mark = 95

```
def mm(a):
    if len(a)==1:
        return a[0],a[0]
    m=len(a)//2
    x1,y1=mm(a[:m])
    x2,y2=mm(a[m:])
    return min(x1,x2),max(y1,y2)
n=int(input())
a=list(map(int,input().split()))
mn,mx=mm(a)
print("Minimum Mark =",mn)
print("Maximum Mark =",mx)
```

```
8
78 92 65 88 95 72 81 69
Minimum Mark = 65
Maximum Mark = 95
```

Question 8: Stock Price Analysis

An investor wants to know the minimum and maximum stock prices recorded during a trading week.

Input

6
450 520 480 610 430 590

Expected Output

Minimum Stock Price = 430
Maximum Stock Price =
610

```python
def mm(a):
    if len(a)==1:
        return a[0],a[0]
    m=len(a)//2
    x1,y1=mm(a[:m])
    x2,y2=mm(a[m:])
    return min(x1,x2),max(y1,y2)
n=int(input())
a=list(map(int,input().split()))
mn,mx=mm(a)
print("Minimum Stock Price =",mn)
print("Maximum Stock Price =",mx)
```

```
6
450 520 480 610 430 590
Minimum Stock Price = 430
Maximum Stock Price = 610
```

Question 9: Fastest and Slowest Race Times

A sports analyst needs to find the fastest (minimum) and slowest (maximum) race completion times.

Input

9
 14 12 18 11 15 13 17 16 10

Expected Output

Fastest Time = 10
 Slowest Time = 18

```python
def mm(a):
    if len(a)==1:
        return a[0],a[0]
    m=len(a)//2
    x1,y1=mm(a[:m])
    x2,y2=mm(a[m:])
    return min(x1,x2),max(y1,y2)
n=int(input())
a=list(map(int,input().split()))
mn,mx=mm(a)
print("Fastest Time =",mn)
print("Slowest Time =",mx)
```

```
9
14 12 18 11 15 13 17 16 10
Fastest Time = 10
Slowest Time = 18
```

Question 10: Largest and Smallest Building Heights

An urban planner is analyzing building heights in a city block and wants to determine the tallest and shortest buildings.

Input

10
 120 150 98 175 140 165 110 190 130 145

Expected Output

Minimum Height = 98
 Maximum Height = 190

```python
def mm(a):
    if len(a)==1:
        return a[0],a[0]
    m=len(a)//2
    x1,y1=mm(a[:m])
    x2,y2=mm(a[m:])
    return min(x1,x2),max(y1,y2)
n=int(input())
a=list(map(int,input().split()))
mn,mx=mm(a)
print("Minimum Height =",mn)
print("Maximum Height =",mx)
```

```
10
120 150 98 175 140 165 110 190 130 145
Minimum Height = 98
Maximum Height = 190
```